

A REPRODUCIBLE REMOTE-SENSING PILOT

# FirePA

## A Reproducible Remote-Sensing Pilot for Provisional Thermal Events in Coclé, Panama

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FIREPA SCIENTIFIC PILOT V1

Jose Rodriguez

Study window

1 January - 30 April 2025

Scientific freeze

7694da7df5808911de48de84016759c5fd22f176

Public website

<https://firepa.pages.dev/>

Source repository

<https://github.com/jrod-md/FirePA>

PUBLICATION / AUGUST 2026

## SECTION 00

# Abstract

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## SCOPE, METHOD, AND EVIDENCE BOUNDARY

FirePA evaluates how satellite thermal detections can be reduced into reproducible provisional analytical units and compared with descriptive optical and external evidence without converting those signals into confirmed wildfire claims. The pilot audits 1,532 NASA FIRMS raw detections, retains 1,185 valid detections inside Coclé, and groups them with the frozen `r1500_t06` spatiotemporal connected-components configuration. The result is 611 provisional thermal events: 344 singletons, 267 multi-detection events, and 17 cases flagged `possible_chain_merge`.

A frozen Sentinel-2 cohort contains 30 events. Twenty-eight are observable under the optical policy and two remain unobserved; those two are not negative cases. NBR and dNBR are retained as descriptive evidence only. Formal human observations equal zero, no ground truth exists, and no supervised model or severity classification was produced.

Two external-reference comparisons preserve an inclusive 5,000 m rule. Cerro Guacamaya has six matched provisional clusters separated by five gaps greater than six hours over 72.733 hours, supporting `fragmentation_possible = true` while preserving all six clusters separately. Cerro Los Picachos has zero official matches; the nearest documented contemporary signal is 10,400.826 m away, so the result is `SPATIAL_THRESHOLD_MISS`, not absence of fire, activity, or thermal signal.

The pilot is static, descriptive, and reproducible within its public boundary. It is not real-time, operational, predictive, or validated ground truth.

**Keywords**

remote sensing / thermal anomalies / NASA FIRMS / Sentinel-2 / spatiotemporal grouping / reproducible research / Coclé, Panama

## SECTION 00

# Resumen

## ALCANCE, MÉTODO Y LÍMITE DE LA EVIDENCIA

FirePA evalúa cómo las detecciones térmicas satelitales pueden reducirse a unidades analíticas provisionales y reproducibles, y compararse con evidencia óptica y referencias externas descriptivas sin convertir esas señales en incendios forestales confirmados. El piloto audita 1.532 detecciones crudas de NASA FIRMS, conserva 1.185 detecciones válidas dentro de Coclé y las agrupa mediante la configuración congelada de componentes conexos espaciotemporales `r1500_t06`. El resultado son 611 eventos térmicos provisionales: 344 eventos de una sola detección, 267 eventos de múltiples detecciones y 17 casos marcados con la bandera técnica `possible_chain_merge`.

Una cohorte congelada de Sentinel-2 contiene 30 eventos. Veintiocho son observables según la política óptica y dos permanecen sin observación óptica bajo esa política (estado técnico: `unobserved`); esos dos casos no son negativos. NBR y dNBR se conservan solo como evidencia descriptiva. Las observaciones humanas formales son cero, no existe verdad de terreno y no se produjo ningún modelo supervisado ni clasificación de severidad validada.

Dos comparaciones con referencias externas conservan una regla inclusiva de 5.000 m. Cerro Guacamaya presenta seis conglomerados provisionales emparejados, separados por cinco intervalos mayores de seis horas a lo largo de 72,733 horas. Esto respalda la interpretación registrada como `fragmentation_possible = true`, manteniendo separados los seis conglomerados. Cerro Los Picachos presenta cero coincidencias oficiales; la señal contemporánea documentada más cercana está a 10.400,826 m, por lo que el diagnóstico técnico registrado es `SPATIAL_THRESHOLD_MISS`; no implica ausencia de incendio, actividad o señal térmica.

El piloto es estático, descriptivo y reproducible dentro de su límite público. No es un sistema en tiempo real, operativo, predictivo ni validado como verdad de terreno.

## Términos clave

teledetección / anomalías térmicas / NASA FIRMS / Sentinel-2 / agrupamiento espaciotemporal / investigación reproducible / Coclé, Panamá

SECTION 00

# Key results and study frame

THE FROZEN ANALYTICAL RECORD AT A GLANCE

1,532

AUDITED FIRMS RAW DETECTIONS

1,185

PROCESSED DETECTIONS IN COCLÉ

611

PROVISIONAL THERMAL EVENTS

r1500\_t06

1,500 m / 6 h

FROZEN CONFIGURATION

30 / 28 / 2

selected / observable / unobserved

OPTICAL COHORT

7 / 2

documents / incidents

EXTERNAL REGISTRY

EVIDENCE BOUNDARY

No ground truth / no predictive model / not operational

The values above are frozen records, processed detections, provisional groups, and descriptive cohorts. They do not form a sequence of confirmed wildfire counts.

Scientific unit: provisional thermal event. Study area: Coclé, Panama. Study window: 2025-01-01 through 2025-04-30, inclusive, UTC.

## SECTION 01

# Research question and objectives

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## A QUESTION ABOUT SIGNALS AND THEIR LIMITS

## *Can thermal signals tell us what happened on the ground?*

FirePA treats this as a bounded research question rather than an operational promise. Satellite thermal anomalies provide repeatable observations, but they do not independently establish ignition, cause, perimeter, severity, duration, or incident identity. The pilot therefore studies the evidence boundary as deliberately as it studies the detections.

The work has four objectives:

- audit and spatially filter a historical NASA FIRMS cohort;
- construct reproducible provisional thermal-event units;
- describe a frozen optical follow-up cohort without creating labels; and
- compare frozen units with limited external incident references while exposing threshold and representation failures.

The objective is not to prove event truth, build a classifier, or deliver an alerting system. The contribution is a reproducible way to preserve the distinction between detection, grouping, observation, and interpretation.

## SECTION 02

# Study area and study window

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COCLÉ, PANAMA / 1 JANUARY - 30 APRIL 2025

The study area is the province of Coclé, Panama. The acquisition and processing window is 2025-01-01 through 2025-04-30, inclusive, in UTC. This boundary and period constrain every frozen count and finding in this report.

The validated local administrative boundary originated from the Instituto Geográfico Nacional Tommy Guardia / ANATI province layer `limi_prov_a`, documented at 1:25,000 scale in EPSG:32617. The Coclé geometry was selected locally and transformed to EPSG:4326 for filtering and public presentation.

The original boundary source and generated local `data/reference/cocle.geojson` are not redistributed because the source metadata declares CC BY-NC-SA. The public website uses a tracked deterministic presentation derivative when the local source boundary is absent. That derivative is contextual geometry, not a scientific result, a burn perimeter, or synthetic geography.

**Scope boundary**

- Geographic generalization beyond Coclé is not claimed.
- Temporal generalization beyond the frozen four-month window is not claimed.
- Public event coordinates are centroids for inspection, not precision ground observations.

SECTION 03

# Data sources and provenance

SCIENTIFIC INPUTS SEPARATED FROM RELATIONAL REFERENCES

FirePA maintains a provenance boundary between scientific inputs, derived public artifacts, and external incident references. A source URL documents provenance; it does not transfer licensing rights and does not become a truth label.

| Source                    | Role in FirePA                                         | Boundary                            |
|---------------------------|--------------------------------------------------------|-------------------------------------|
| NASA FIRMS / LANCE        | VIIRS thermal-anomaly detections for the frozen cohort | Detection is not confirmed wildfire |
| Sentinel-2 SR Harmonized  | Descriptive optical follow-up for 30 frozen events     | No severity class or truth label    |
| Cloud Score+              | Optical clarity assessment for linked scenes           | Clarity is not evidence of fire     |
| IGN Tommy Guardia / ANATI | Validated Coclé administrative boundary                | Original source not redistributed   |
| 7 external documents      | Relational context for two documented incidents        | Not pipeline inputs or ground truth |

External references

The external registry contains seven manually located documents describing Cerro Los Picachos and Cerro Guacamaya. Publisher claims about dates, affected area, and cause are preserved per source rather than adjudicated into a single value. Matching coordinates are approximate researcher-provided anchors; FirePA does not claim that the source articles supplied those coordinates.

The manual search was not exhaustive. No qualifying articles were located in March or April 2025, but absence of a located article is not absence of fire, activity, or thermal signal.

SECTION 04

# FIRMS acquisition and audit

FROM 1,532 RAW RECORDS TO 1,185 DETECTIONS IN COCLÉ



FROZEN REDUCTION / RECORDS -> DETECTIONS -> ANALYTICAL GROUPS

| Audit stage                            | Count |
|----------------------------------------|-------|
| Audited raw detections                 | 1,532 |
| Valid schema/date/coordinate records   | 1,532 |
| Detections inside the requested period | 1,532 |
| Processed detections inside Coclé      | 1,185 |

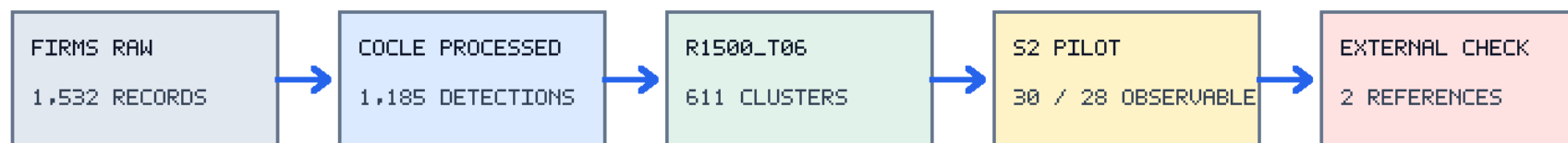
The local workflow used NASA FIRMS VIIRS historical detections. It checked source availability, split requests into bounded temporal fragments, retained raw bytes and manifests locally, validated required fields, and filtered records by time and the validated Coclé geometry.

The difference between 1,532 and 1,185 is the spatial study-area filter. FIRMS rows are satellite-observed thermal anomalies. Multiple rows may describe the same analytical event, and a single row is not automatically a confirmed fire.

## FIREPA PILOT: LOCAL SCIENTIFIC PIPELINE

FROZEN INPUTS AND DESCRIPTIVE OUTPUTS ? COUNTS ARE RECORDS OR PROVISIONAL CLUSTERS

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FIXED PERIOD: 2025-01-01 TO 2025-04-30 ? AOI: COCLE ? NO RECLUSTERING IN THIS REPORT

OFFICIAL EXTERNAL RULE:  $\leq 5$  KM + INCLUSIVE DATE OVERLAP

FINAL PHASE: LOCAL READ-ONLY ANALYSIS ? NO NEW NETWORK OR EARTH ENGINE CALLS

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PROVENANCE: LOCAL FIREPA CHECKOUT ? STATIC PNG ? DESCRIPTIVE, NOT A DASHBOARD OR OPERATIONAL VIEW

**Figure 1. Frozen scientific pipeline.** The byte-frozen pipeline plate records the local descriptive sequence and the official external-reference rule. It is reproduced proportionally from `site-data/figures/01_pipeline_overview.png` without pixel-level modification.

## SECTION 05

# Spatiotemporal grouping

## CONNECTED COMPONENTS UNDER R1500\_T06

The frozen configuration `r1500_t06` applies connected components with a spatial link distance of 1,500 m or less, a temporal link difference of six hours or less, and metric distance calculations in EPSG:32617. Links use coordinates and UTC timestamps only.

| Frozen result              | Count |
|----------------------------|-------|
| Provisional thermal events | 611   |
| Singleton events           | 344   |
| Multi-detection events     | 267   |
| possible_chain_merge       | 17    |

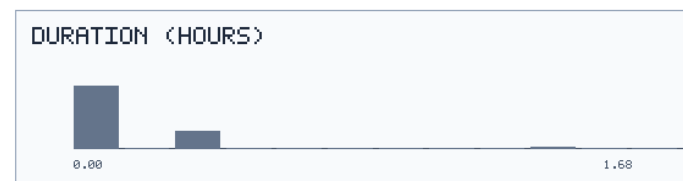
FRP, confidence, source, satellite, instrument, and day/night status remain descriptive attributes and do not create graph edges. A provisional event is therefore a connected analytical group, not a claim of one ignition, one continuous incident, a burn-scar perimeter, or a confirmed wildfire.

Connected components can join observations through chains or split prolonged activity when temporal gaps exceed six hours. The `possible_chain_merge` field is a review flag, not a corrected class or an automatic merge instruction.

*The term provisional is essential: the pilot has no ground-truth adjudication.*

## R1500\_T06 CLUSTER DISTRIBUTION

N = 611 PROVISIONAL CLUSTERS ? GUACAMAYA SUMMARY IS REPORTED SEPARATELY



SOURCE: OUTPUTS/CLUSTERING/R1500\_T06/EVENTS.CSV ? HISTOGRAMS ARE DESCRIPTIVE? CLUSTERS ARE NOT CONFIRMED FIRES

**Figure 2. Descriptive cluster distribution.** Frozen descriptive histograms for the 611 r1500\_t06 clusters. The figure does not establish confirmed-fire classes or infer severity. Source artifact: site-data/figures/02\_cluster\_distribution.png.

## SECTION 06

# Sentinel-2 follow-up

## A FROZEN 30-EVENT OPTICAL COHORT

The structural optical cohort contains 30 frozen events. Sentinel-2 Surface Reflectance Harmonized and Cloud Score+ were used under the documented observability policy. Twenty-eight events are observable and two remain unobserved.

Unobserved means that the policy did not obtain usable optical support. It does not mean no fire, no event, or no land-surface change. The two cases are not negative examples.

**Evidence modes remain separate**

- `selected_pair`: one selected pre/post pair under the frozen policy;
- `window_median`: a separate seven-case technical pilot using local numeric bundles and per-band temporal medians.

The window-median pilot passed technical integrity checks but was never connected to formal review assignments and does not replace selected-pair evidence.

## SECTION 07

## NBR / dNBR evidence boundary

```
NBR = (B8 - B12) / (B8 + B12)
dNBR = NBR_pre - NBR_post
```

B8 and B12 are evaluated on common 20 m support. These indices are descriptive and may respond to cloud, haze, phenology, moisture, agriculture, exposed soil, water, and other land-surface changes. FirePA produced no `significant_burn` target, validated burn-scar label, severity class, or causal attribution.

## SECTION 08

# Review and model-assisted calibration

## EXPLORATORY INSTRUMENT TESTING, NOT VALIDATION

```
execution_authorized = false
formal_human_observations = 0
```

Formal review infrastructure was prepared, but execution was deferred because a qualified reviewer was unavailable. No supervised predictive model was trained because the pilot has no defensible target or ground truth.

A blind seven-case model-assisted calibration tested whether a proposed two-pass review contract preserved ambiguity, confidence, temporal robustness, observation limitations, and competing land-surface explanations. Pass A examined selected multispectral pre/post evidence. Pass B examined a separate temporal-robustness view while preserving the Pass A fields. Responses were recorded as `provisional_pseudolabel`.

A later control used the same seven anonymized cases with two independently returned model-assisted reviews. The comparison contained 14 provisional rows. There was no case with exact agreement across every structured field. Agreement between models from the same family is not independence from a reference standard and is not an accuracy measure.

The calibration affected none of the 1,532 raw records, 1,185 processed detections, 611 provisional events, 30-event optical cohort, external-reference results, frozen figures, or public-package artifacts. It produced no human observation, ground truth, training target, confirmed-fire label, severity class, model-performance estimate, or predictive model.

Public reproducibility is limited to the aggregate methodological record. The case-to-event mapping, response JSON, review database, source panels, local review packages, and protected optical inputs are not redistributed.

SECTION 09

# External-reference methodology

RELATIONAL EVIDENCE UNDER A FIXED INCLUSIVE RULE

The frozen registry contains seven documents describing two incidents. Each source preserves its own publication date, event date, location, area, and cause claim where available. FirePA does not adjudicate a single true area or cause.

The comparison reads the 611 frozen event centroids without reclustering. A match must satisfy both conditions:

- inclusive temporal overlap with the incident reference window; and
- Euclidean metric distance of 5,000 m or less from an approximate reference anchor in EPSG:32617.

The two matching anchors were supplied by the researcher and are explicitly approximate. They are not claimed to have been published by the articles. The reference URLs are provenance records only; they were not inputs to the FIRMS pipeline, clustering, optical analysis, or modeling.

Official frozen comparisons

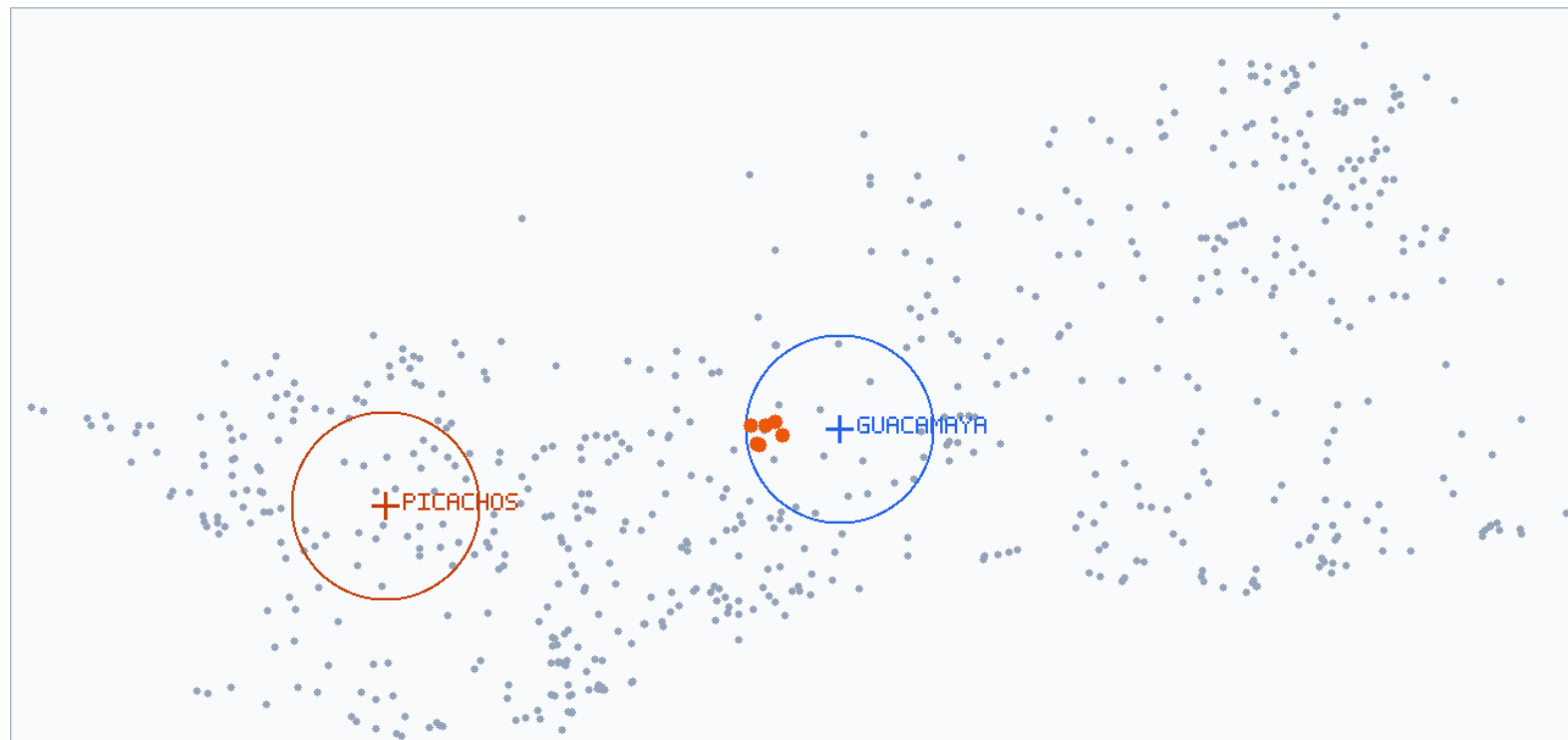
| Reference          | Window         | Official result          |
|--------------------|----------------|--------------------------|
| Cerro Guacamaya    | 23-27 Jan 2025 | 6 matches within 5,000 m |
| Cerro Los Picachos | 15-17 Jan 2025 | 0 matches within 5,000 m |

*These comparisons are relational evidence, not ground truth, institutional validation, labels, or accuracy tests.*

## EXTERNAL REFERENCE CHECK: SPATIAL CONTEXT

611 FROZEN CLUSTER CENTROIDS ? 5 KM RINGS ARE VISUAL APPROXIMATIONS

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RINGS: 5 KM APPROXIMATION FOR DISPLAY ? OFFICIAL DISTANCES USE EPSG:32617 EUCLIDEAN METERS

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SOURCE: FROZEN CLUSTER CENTROIDS, EXTERNAL\_REFERENCE\_CHECK\_V1, LOCAL DISTANCE FUNCTION ? RELATIONSHIP CHECK ONLY

**Figure 3. External-reference spatial context.** Frozen spatial context for 611 event centroids and the two approximate reference anchors. The displayed 5 km rings are visual approximations; official distances use Euclidean meters in EPSG:32617. Source artifact: site-data/figures/03\_external\_reference\_map.png.

SECTION 10

# Case study: Cerro Guacamaya

SIX CLUSTERS, FIVE GAPS, ONE REPRESENTATION WARNING

The official 2025-01-23 to 2025-01-27 comparison returns six matched provisional clusters within the inclusive 5,000 m radius. They contain 2, 4, 3, 5, 1, and 4 detections. One cluster is flagged `possible_chain_merge`; that review flag does not change the six-unit result.

| Cluster | Detections | Gap from prior (h) | Max FRP (MW) |
|---------|------------|--------------------|--------------|
| C1      | 2          | -                  | 7.69         |
| C2      | 4          | 12.383             | 5.05         |
| C3      | 3          | 23.300             | 13.67        |
| C4      | 5          | 10.933             | 7.92         |
| C5      | 1          | 11.083             | 4.49         |
| C6      | 4          | 12.617             | 2.91         |

The first cluster begins at 2025-01-24 06:11Z and the last ends at 2025-01-27 06:55Z, producing a 72.733 h overall span. The five inter-cluster gaps are 12.383 h, 23.300 h, 10.933 h, 11.083 h, and 12.617 h. Every gap exceeds the frozen six-hour link window.

The frozen interpretation is `fragmentation_possible = true`. A prolonged documented incident may correspond to multiple FirePA events. This is a representation limitation, not proof that the clustering is wrong and not authorization to merge the six provisional units into one continuous spreading fire.

## CERRO GUACAMAYA: MATCHED CLUSTER TIMELINE

SIX OFFICIAL MATCHES ? PROLONGED GAPS ARE SHOWN AGAINST THE FROZEN 6 HOUR WINDOW



GREEN = DAY ? PURPLE = NIGHT ? RED LABELS = INTER-CLUSTER GAP

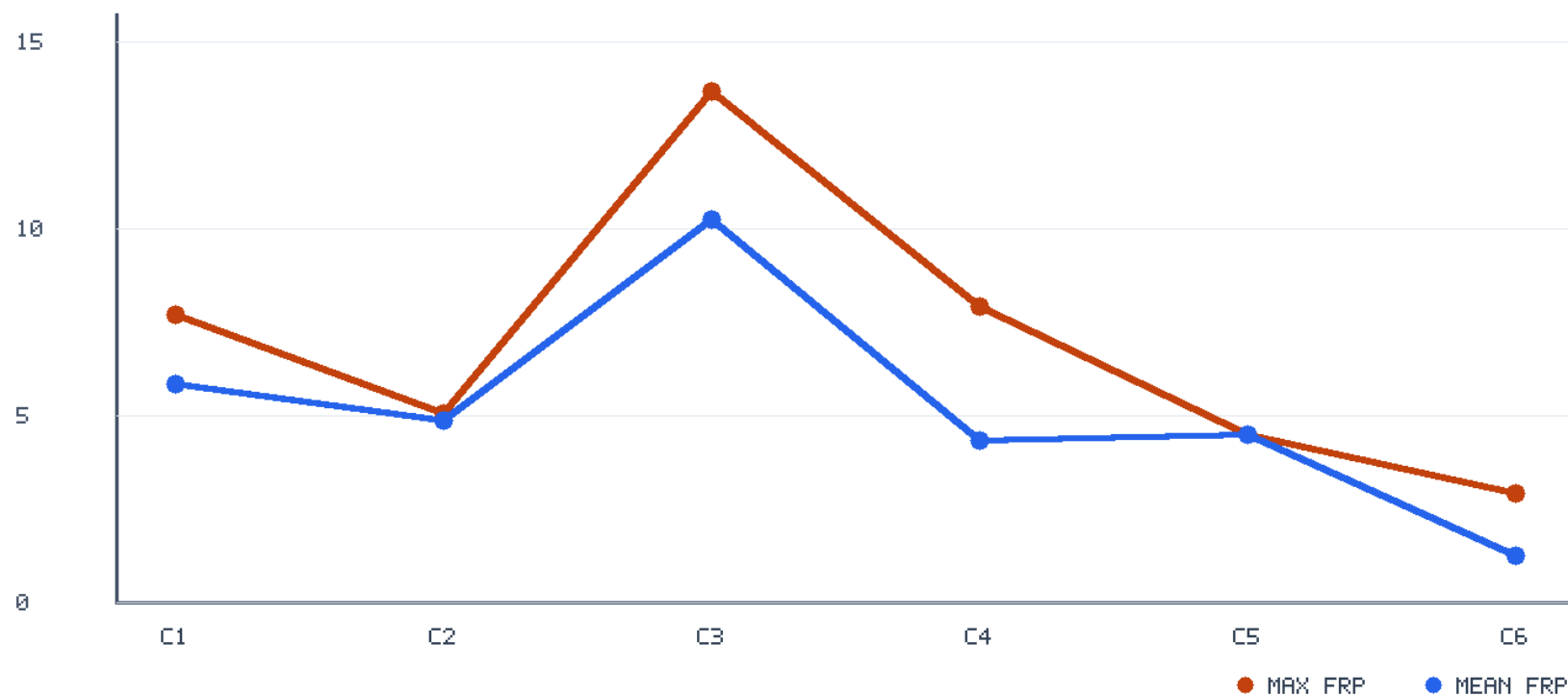
THE GAP PATTERN IS A REPRESENTATION LIMITATION UNDER T06, NOT AN AUTOMATIC BUG FINDING

SOURCE: OFFICIAL GUACAMAYA MATCHES + R1500.T06 MEMBERSHIP.CSV ? STATIC CHRONOLOGY IN UTC

**Figure 4. Cerro Guacamaya matched-cluster timeline.** Six official matches are shown separately against the frozen six-hour window. Red labels mark inter-cluster gaps. The gap pattern is a representation limitation under t06, not an automatic merge or confirmed continuous incident. Source artifact: site-data/figures/04\_guacamaya\_timeline.png.

## CERRO GUACAMAYA: DESCRIPTIVE FRP SEQUENCE

MAXIMUM AND MEAN FRP BY CHRONOLOGICAL MATCHED CLUSTER ? MW



DESCRIPTIVE SEQUENCE ONLY? NO COMPOSITE SCORE OR INFERENCE FROM FRP

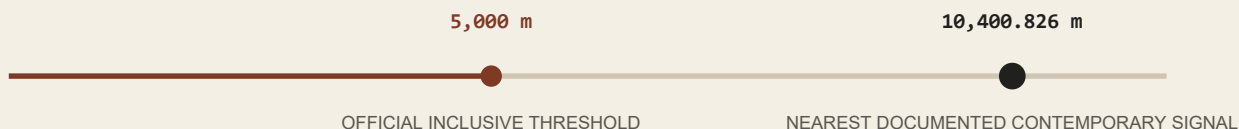
SOURCE: SIX OFFICIAL GUACAMAYA MATCHES AND CLUSTER MEMBERSHIP ? CHRONOLOGICAL ORDER C1-C6

**Figure 5. Cerro Guacamaya descriptive FRP sequence.** Maximum and mean FRP are shown by chronological matched cluster. FRP remains descriptive; the sequence is not a composite score, severity estimate, or causal inference. Source artifact: site-data/figures/05\_guacamaya\_frp\_distribution.png.

## SECTION 11

# Case study: Cerro Los Picachos

A THRESHOLD-SPECIFIC ZERO, NOT ZERO SIGNAL



## SPATIAL\_THRESHOLD\_MISS

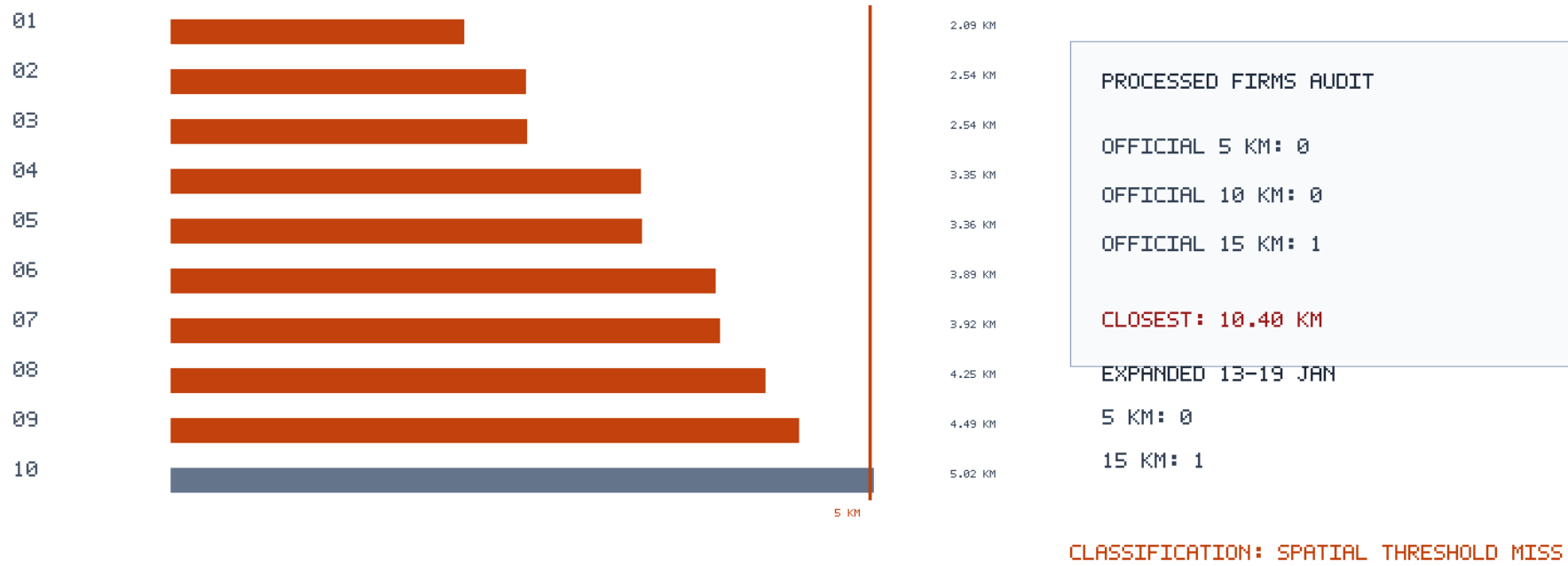
The official 2025-01-15 to 2025-01-17 comparison returns zero matches within the inclusive 5,000 m radius. The nearest documented contemporary frozen event is 10,400.826 m from the approximate anchor. The correct frozen diagnosis is SPATIAL\_THRESHOLD\_MISS.

The processed-detection audit records zero detections within 5 km, zero within 10 km, and one within 15 km. Extending the diagnostic date window to 2025-01-13 through 2025-01-19 does not change the 5 km result. These 10 km and 15 km checks are diagnostic context only; they do not alter the official 5,000 m rule.

*Zero at 5 km must not be translated into no fire, no activity, no thermal signal, or absence of an event. It states only that no frozen event met the official temporal-overlap and distance rule.*

# CERRO LOS PICACHOS: DIAGNOSTIC OF OFFICIAL ZERO

OFFICIAL WINDOW 2025-01-15 TO 2025-01-17 ? 5 KM RULE PRESERVED



SOURCE: 1,185 PROCESSED FIRMS DETECTIONS + 611 FROZEN CLUSTERS ? 0 AT 5 KM IS NOT 0 SIGNAL IN THE REGION

**Figure 6. Cerro Los Picachos diagnostic of the official zero.** The frozen plate preserves the official 5 km rule and shows the nearest documented contemporary signal at approximately 10.40 km. Wider-distance checks remain diagnostic context, not a changed matching rule. Source artifact: site-data/figures/06\_los\_picachos\_diagnostic.png.

## SECTION 12

# Scientific and interpretive limitations

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## THE INFERENCE BOUNDARY IS A PRIMARY RESULT

FirePA is a descriptive research pilot. Its restraint is methodological, not rhetorical: each analytical stage records what the evidence supports and where interpretation must stop.

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- A FIRMS anomaly does not independently confirm a wildfire.
- A provisional event is an analytical group, not necessarily one physical incident.
- An event centroid is not a burn perimeter or precision ground location.
- An unobserved optical case is not a negative case.
- NBR and dNBR are descriptive indices, not validated severity.
- Formal human observations equal zero.
- No ground truth or institutional validation exists.
- No accuracy estimate, supervised predictive model, or generalization claim exists.
- The external-reference search is limited and non-exhaustive.
- Guacamaya may expose temporal fragmentation under the six-hour convention.
- The Picachos zero is specific to the inclusive 5,000 m rule.
- Findings are limited to Coclé and the frozen study period.

*FirePA does not provide real-time monitoring, alerts, emergency guidance, forecasting, validated decision support, causal attribution, or operational wildfire classification.*

## SECTION 13

# Reproducibility and public package

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## A DETERMINISTIC PACKAGE WITH AN EXPLICIT PROTECTED-SOURCE BOUNDARY

The machine-readable `site-data/` package is the highest public authority. It contains exactly 611 WGS84 event centroids, two approximate reference anchors, the Guacamaya chronology, methodology, citations, provenance, six byte-frozen figures, and a manifest SHA-256 ledger.

A clean clone can build and test the bilingual Astro publication and verify the complete public package without a backend, network access, Earth Engine, or the non-redistributed Coclé source boundary.

```
cd site
npm ci
npm run build
npm run test

python scripts/verify_public_release_package.py --package site-data
python -m pytest -q
```

A clean clone cannot fully regenerate the scientific history. Raw FIRMS rows and local manifests, processed/interim research tables, Sentinel-2 rasters and Earth Engine outputs, the original administrative boundary, private review mappings and databases, historical model-assisted responses, and generated review bundles are intentionally excluded.

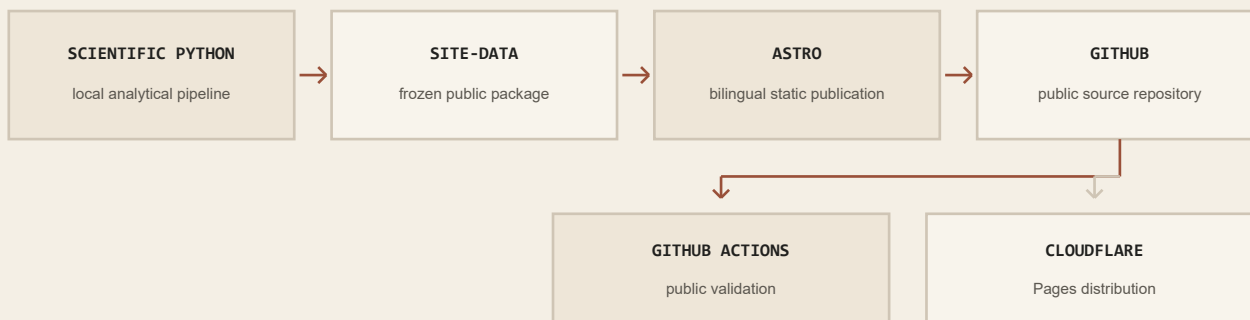
When every protected artifact is available, the explicit full-source gate verifies their hashes. It is expected to fail in a clean public clone rather than fabricate substitutes:

```
python scripts/verify_public_release_package.py --package site-data --full-source
```

## SECTION 14

# Software and publication architecture

## CURRENT INFRASTRUCTURE, SEPARATED FROM SCIENTIFIC EVIDENCE



GitHub Actions verifies the clean-clone public contract.

Cloudflare Pages distributes the publication; neither service is scientific evidence.

The analytical implementation is a local Python package with verification tests. Its approved outputs are projected into the frontend-agnostic `site-data/` package. The public publication is a bilingual static Astro site in English and Latin American Spanish. Source is hosted in the public GitHub repository and the static site is distributed through Cloudflare Pages.

The website can be built from the public package without private inputs. Its Coclé presentation geometry falls back only to a tracked deterministic derivative; it does not invent replacement cartography.

On pushes and pull requests targeting `main`, GitHub Actions runs two clean-clone jobs: the public package verifier and public Python test suite; and `npm ci`, the Astro build, and the Astro publication contract. It does not run the `--full-source` gate or validate protected/private inputs. Cloudflare Pages publishes the static site from the public repository; it is distribution infrastructure, not scientific evidence or an operational wildfire platform.

### Public endpoints

Website: <https://firepa.pages.dev/>

Repository: <https://github.com/jrod-md/FirePA>

## SECTION 15

# Conclusion

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**A BOUNDED EVIDENCE STRUCTURE, NOT A CLASSIFIER**

FirePA demonstrates a reproducible reduction from satellite thermal signals to provisional analytical events while preserving uncertainty. It audits 1,532 FIRMS detections, retains 1,185 records in the study area, and groups them into 611 provisional events under a frozen and inspectable convention. Optical follow-up and external references add context without becoming labels.

The two case studies expose complementary limits. Cerro Guacamaya shows that prolonged documented activity can align with six separate groups when every inter-cluster gap exceeds the six-hour link window. Cerro Los Picachos shows that zero matches within an official 5 km threshold can coexist with a documented contemporary signal 10,400.826 m from the approximate anchor.

Neither case provides ground truth. Together, they show why threshold results and analytical units must be reported in the language of the procedure that produced them.

*The principal contribution is not a wildfire classifier.  
It is a bounded evidence structure.*

Within that structure, provisional units remain provisional, unobserved cases remain unresolved, optical indices remain descriptive, model-assisted calibration remains exploratory, and external references remain relational. This restraint makes the publication reproducible without claiming more than its sources can support.

## SECTION R

# References

## SCIENTIFIC SOURCES AND EXTERNAL INCIDENT DOCUMENTS

- [S1] NASA FIRMS / LANCE. Area API, data availability service, and API documentation.  
[https://firms.modaps.eosdis.nasa.gov/content/academy/data\\_api/firms\\_api\\_use.html](https://firms.modaps.eosdis.nasa.gov/content/academy/data_api/firms_api_use.html)
- [S2] Copernicus Sentinel-2 Surface Reflectance Harmonized. Google Earth Engine data catalog.  
[https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS\\_S2\\_SR\\_HARMONIZED](https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED)
- [S3] Google Cloud Score+ S2 Harmonized. Google Earth Engine data catalog.  
[https://developers.google.com/earth-engine/datasets/catalog/GOOGLE\\_CLOUD\\_SCORE\\_PLUS\\_V1\\_S2\\_HARMONIZED](https://developers.google.com/earth-engine/datasets/catalog/GOOGLE_CLOUD_SCORE_PLUS_V1_S2_HARMONIZED)
- [S4] Instituto Geográfico Nacional Tommy Guardia / ANATI. Administrative divisions source page.  
<https://ignpanama.anati.gob.pa/index.php/divisionpolitica-administrativa>

## External incident source registry

- [E001] TVN Noticias. 2025-01-17. Cerro Los Picachos.  
[https://www.tvn-2.com/nacionales/incendio-picachos-ola-afecto-extensa-area-investigan-causas\\_1\\_2173253.html](https://www.tvn-2.com/nacionales/incendio-picachos-ola-afecto-extensa-area-investigan-causas_1_2173253.html)
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## Reference boundary

No missing authors, DOIs, issue numbers, access dates, venues, or publication metadata have been inferred. The seven external URLs are provenance records and relational context, not scientific truth labels.

SECTION A

# Public package inventory

COMPACT FILE AND PURPOSE REGISTER

Package version: firepa-public-release-v1. Scientific freeze: 7694da7df5808911de48de84016759c5fd22f176.

| Path                                      | Purpose                                               |
|-------------------------------------------|-------------------------------------------------------|
| project-summary.json                      | Frozen counts, scope, period, and findings            |
| events.geojson                            | Exactly 611 public WGS84 event centroids              |
| external-references.geojson               | Two approximate comparison anchors                    |
| guacamaya-timeline.json                   | Six-cluster chronology and five gaps                  |
| methodology.json                          | Machine-readable workflow and limits                  |
| citations.json                            | Scientific-document and external-source registry      |
| provenance.json                           | Inputs, transformations, hashes, and privacy boundary |
| figures/01_pipeline_overview.png          | Byte-frozen scientific pipeline plate                 |
| figures/02_cluster_distribution.png       | Byte-frozen descriptive cluster distributions         |
| figures/03_external_reference_map.png     | Byte-frozen external-reference context                |
| figures/04_guacamaya_timeline.png         | Byte-frozen Guacamaya chronology                      |
| figures/05_guacamaya_frp_distribution.png | Byte-frozen Guacamaya FRP description                 |
| figures/06_los_picachos_diagnostic.png    | Byte-frozen Picachos threshold diagnostic             |
| manifest.json                             | Deterministic public-package SHA-256 ledger           |

Public event properties are descriptive. None is a confirmed-fire, severity, ground-truth, or predictive-model field.

SECTION B

# Frozen figure integrity

DIMENSIONS AND SHA-256 LEDGER

All six figures are public derivatives copied byte-for-byte from the frozen final scientific report. Their scientific content was not recomputed. They were placed proportionally in this PDF without cropping, recoloring, relabeling, sharpening, or destructive resizing.

| Figure                            | Dimensions | SHA-256                                                          |
|-----------------------------------|------------|------------------------------------------------------------------|
| 01_pipeline_overview.png          | 1600 x 900 | f818cc2f0ebe93a450c6dff5c505849a91c71bf013e597bdc6852a8cc7b94d1c |
| 02_cluster_distribution.png       | 1600 x 900 | 5941c237f74a4ad95809aa7a0430a70269337560aa4cbabc62601d03edf2c2ea |
| 03_external_reference_map.png     | 1600 x 900 | 3f0bcd569a0d762bb1c2435fe25158bf9ecbdebb1ada682c04c1a299a32916e7 |
| 04_guacamaya_timeline.png         | 1600 x 900 | d248bdda6b88eee52dc27f2567a29ec3bcf494a72b139ffc1340bed7db4566d0 |
| 05_guacamaya_frp_distribution.png | 1600 x 900 | 973ef4f72e0f15873a11007359a6dba5aae419d4b83a78a855f428227a4c777b |
| 06_los_picachos_diagnostic.png    | 1600 x 900 | 8fa486e16e97063ad5f55b76165a3c671c2b1acd6b6c5c717a838ef56d4176a2 |

Integrity rule

The generation script verifies every source PNG against the public provenance ledger before creating the PDF. Publication QA repeats the check after generation. A mismatch blocks report generation.

## SECTION C

# Reproducibility commands

---

## CLEAN-CLONE VERIFICATION AND PROTECTED-SOURCE BOUNDARY

**Website verification**

```
cd site
npm ci
npm run build
npm run test
```

The Astro build creates the English and Latin American Spanish static routes. The site contract verifies the public data and all six frozen figures.

**Public Python verification**

```
python scripts/verify_public_release_package.py --package site-data
python -m pytest -q
```

The default verifier checks the deterministic public package. Self-contained tests run; tests whose named protected inputs are absent skip explicitly.

**Full-source gate**

```
python scripts/verify_public_release_package.py --package site-data --full-source
```

This stricter mode requires every excluded source artifact and compares its frozen hash. It is expected to fail in a clean public clone. The pipeline must fail rather than fabricate a scientific substitute.

---

**Report regeneration**

```
python report/build_firepa_report.py
```

The report source reads only the public package, the citation author, and the six frozen PNGs. It does not recalculate events, distances, optical metrics, or scientific figures.